

DEEP RABADIYA

Ahmedabad

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in LinkedIn

📁 Portfolio

Professional Summary

Agriculture professional with a B.Sc. (Hons.) in Agriculture and M.Sc. in Agriculture Analytics, combining comprehensive agricultural knowledge with data-driven approaches to crop production, crop monitoring, soil analysis, yield estimation, agricultural risk analysis, and farm decision-making. Academic training includes agronomy, crop production, crop physiology, plant breeding, genetics, plant pathology, entomology, horticulture, soil science, irrigation and water management, integrated pest management, agricultural economics, farm management, protected cultivation, sustainable agriculture, organic farming, biofertilizers, seed technology, and post-harvest management. Experienced in applying Machine Learning, statistical modelling, weather and water analytics, agricultural datasets, and satellite-derived information to crop classification, crop health assessment, acreage estimation, yield forecasting, soil analysis, and agricultural decision support.

Core Agricultural Skills

Crop Science & Agronomy: Crop Production, Agronomy, Crop Physiology, Crop Management, Field Crop Production, Crop Health Monitoring, Crop Yield Estimation, Crop Yield Forecasting, Crop Classification

Soil, Water & Natural Resources: Soil Science, Soil Fertility Management, Problematic Soil Management, Soil & Water Conservation, Irrigation & Water Management, Soil Moisture Analysis, Soil Property Estimation, Weather & Water Analytics

Plant Protection: Plant Pathology, Plant Disease Detection, Entomology, Integrated Pest Management, Insect Pest Management, Disease Management

Horticulture & Crop Production: Fruit & Plantation Crops, Vegetable & Spice Crops, Ornamental Crops, Protected Cultivation, Post-Harvest Management, Seed Technology

Sustainable Agriculture: Sustainable Agriculture, Farming Systems, Organic Farming, Biofertilizers, Fertilizers & Manures, Farm Resource Management

Agricultural Analytics: Agricultural Data Analysis, Agricultural Market Analysis, Agricultural Risk Analysis, Crop & Soil Analysis, Statistical Modelling, Predictive Modelling, Time-Series Forecasting, Machine Learning, Deep Learning

Technical Skills

Programming & Data Analysis: Python, R, SQL, Pandas, NumPy, Matplotlib, Scikit-learn

Machine Learning & AI: Machine Learning, Deep Learning, TensorFlow, Keras, OpenCV, Random Forest, Ensemble Modelling

Agricultural Remote Sensing: Google Earth Engine, Sentinel-1 SAR, Sentinel-2, ERDAS Imagine, SNAP, ArcGIS, QGIS, ENVI

Agricultural Data & Analytics: PostgreSQL, Power BI, Looker Studio, Advanced Excel, Statistical Modelling, Weather Data Analysis

Geospatial Libraries: Rasterio, GDAL, GeoPandas

Weather & Climate Data: IMD, ERA5, NASA POWER, CHIRPS

Professional Experience

Amnex Infotech Pvt Ltd

Ahmedabad

Associate Remote Sensing Analyst (Agriculture)

April 2026 – Present

- Working on agricultural and geospatial data workflows supporting crop classification, crop monitoring, spatial analysis, and data-driven decision support.
- Performing crop classification using satellite datasets and ERDAS Imagine to support agricultural crop identification and monitoring.
- Using Google Earth Engine for satellite data preprocessing, cloud masking, image compositing, temporal analysis, and generation of crop health and vegetation monitoring products.
- Developing workflows for agricultural weather and climate data extraction, processing, and analysis using IMD, ERA5, NASA POWER, and CHIRPS datasets.
- Applying Python, GIS, and data processing techniques for agricultural data analysis, feature extraction, spatial queries, and automated generation of analytical datasets.
- Conducting thematic mapping, spatial analysis, raster-vector processing, and agricultural data preparation using ArcGIS and QGIS.

SBI General Insurance

Remote Sensing and GIS Analyst (Agriculture)

New Delhi

Jan 2025 – March 2026

- Worked on agricultural applications involving crop classification, crop acreage estimation, and crop yield forecasting using Machine Learning, Deep Learning, satellite-derived information, and field-level data.
- Developed AI/ML and Deep Learning models for crop classification, acreage estimation, and yield forecasting to support agricultural assessment and decision-making.
- Used Sentinel-2 optical imagery and Sentinel-1 SAR data for agricultural crop analysis, satellite image preprocessing, and temporal crop monitoring.
- Used Google Earth Engine for satellite data preprocessing, cloud masking, image compositing, and temporal aggregation for agricultural monitoring.
- Developed workflows for extracting and processing agricultural weather information using IMD, NASA POWER, ERA5, and CHIRPS datasets.
- Applied GIS and Remote Sensing techniques using ArcGIS, QGIS, SNAP, and ERDAS Imagine for agricultural mapping, spatial analysis, and satellite-based crop assessment.

Semantic Technologies and Agritech Services Pvt Ltd

Remote Sensing and GIS Analyst (Agriculture)

Pune, Maharashtra

May 2024 – Jul 2024

- Worked on agritech applications involving crop health monitoring and crop acreage estimation using satellite-derived agricultural information.
- Developed Machine Learning models using SAR and optical satellite data for crop health assessment and agricultural area estimation.
- Worked on DPRVI-based crop health assessment methodologies for monitoring agricultural crop conditions.

Education

Junagadh Agricultural University

B.Sc. (Hons.) Agriculture

Junagadh, Gujarat

2019 – 2023

Dhirubhai Ambani University

M.Sc. Agriculture Analytics

Gandhinagar, Gujarat

2023 – 2025

Indian Institute of Remote Sensing - ISRO

M.Sc. Second Semester

Dehradun

2024

Relevant Agricultural Coursework

Soil Science, Agricultural Meteorology & Climate Change, Agricultural Microbiology, Agricultural Informatics, Agronomy, Soil Fertility Management, Soil & Water Conservation, Agricultural Economics, Fruit & Plantation Crops, Plant Breeding, Plant Genetics, Crop Physiology, Vegetable & Spice Crops, Agricultural Marketing, Integrated Pest Management, Insect Pest Management, Crop Production Technology, Weed Management, Seed Technology, Protected Cultivation, Farming Systems & Sustainable Agriculture, Farm Management, Biofertilizers, Organic Farming, Post-Harvest Management.

Agricultural Analytics Coursework

Earth Observation Systems, Analytics & Statistical Methods, Python Programming & SQL, Big Data Analytics, Machine Learning, Programming for Geodata Processing, Spatial Modelling & Data Assimilation, Agricultural Market Analysis, Crop & Soil Analysis, Risk Analysis & Modelling, Weather & Water Analytics.

Projects

Crop Yield Estimation using ML and Semi-Physical Model

- Developed a Machine Learning stacking ensemble model for crop yield estimation using agricultural and field-level information.
- Developed a semi-physical crop yield model integrating satellite-derived biophysical parameters including fAPAR, PAR, water stress, and temperature stress with field-level data.

Crop Recommendation System

- Built a fruit crop recommendation system using PostgresML, Streamlit, and Machine Learning algorithms to recommend suitable crops based on soil nutrients (N, P, K), soil moisture, temperature, and environmental factors.

- Developed predictive models using historical agricultural data to support crop selection, productivity, and sustainable agricultural decision-making.

Satellite-Based Crop Mapping and Area Estimation

- Worked on satellite-based crop mapping and acreage estimation using Machine Learning, satellite imagery, Python, and agricultural datasets.
- Applied crop classification and spatial analysis techniques to identify agricultural crop areas and estimate cultivated area.

Crop Health Monitoring using SAR and Optical Satellite Data

- Developed agricultural crop health monitoring workflows using SAR and optical satellite imagery.
- Applied Machine Learning and DPRVI-based methodologies to assess crop condition and support agricultural monitoring.

Soil Properties and Soil Moisture Estimation

- Applied Machine Learning, Deep Learning, satellite data, and GIS techniques for soil property and soil moisture estimation.
- Analysed soil-related environmental variables to support agricultural and land-management applications.

Tomato Disease Classification using Deep Learning

- Fine-tuned a pretrained VGG16 Deep Learning model to classify 10 tomato leaf disease classes.
- Applied OpenCV-based image preprocessing, background removal, and data augmentation for agricultural disease classification.

Tomato Price Forecasting using Statistical Models

- Forecasted tomato prices using 14 years of historical price and arrival data in Python.
- Applied ARIMA, SARIMA, ARIMAX, SARIMAX, ARCH, GARCH, VAR, and VARMAX models to analyse and forecast agricultural market trends.

Certifications

- Horticulture Entrepreneurship Training – **Centre of Excellence for Protected Cultivation and Precision Farming on Vegetables**
- Training on Beekeeping under NBHM Project – **AMR Dairy, sponsored by BEE Board**
- Crop Yield Modelling using Advance Geospatial Technologies – **Amnex Infotechnologies**
- Application of Geoinformatics in Ecological Studies – **IIRS, ISRO**
- Spatial Analysis with ArcGIS Pro – **Esri India**
- Microwave Data Processing and Applications – **Space Application Centre, ISRO**
- Advance Excel, Power BI and Tableau – **The Pioneer Tech**
- Developing Web-based Geographical Information Systems – **Nascent Infotech**
- Python for Computer Vision with OpenCV and Deep Learning – **Udemy**